

Chapter 1: Introduction

1.1 Background

Mantle melting—driven by decompression, heating, or water release— is a crucial geophysical process in lithosphere evolution and plate tectonics. Volcanism, for instance, arises from mantle melting and melt segregation, producing Earth’s crust and shaping its chemical evolution. Comparable processes may also occur on other terrestrial planets.

However, prevailing estimates of upper mantle temperatures, together the inferred state of stress in the lithosphere, indicate that the asthenosphere is often partially molten ? and Schmeling (2000). Partial melting takes place in the following three key tectonic settings.

Mid-ocean ridges: Upwelling of the convecting mantle beneath mid-ocean ridges leads to decompression melting and generates the oceanic crust.

Hot-spots: Thermal anomalies induce melting beneath hot-spots and generate ocean islands and oceanic plateaus as well as flood basalt provinces on the continents.

Subduction zones: The addition of fluids that lower the melting point induces flux-melting above the dehydrating down-going oceanic crust and generates the continental crust.

A thorough understanding of partial melting is fundamental to comprehending the dynamics of mantle convection. However, modeling partial melting in the mantle poses significant challenges, as porosity vanishes in melt-free regions, rendering the governing two-phase flow equations highly degenerate. In this thesis, we develop and apply advanced numerical schemes capable of robustly handling such degeneracies to study the temporal dynamics of a partially molten mantle. The simulations resolve

the coupled evolution of porosity and phase transitions, providing a consistent framework for analyzing the thermomechanical processes associated with partial melting.

1.2 Related Work

We review some existing software packages that address related aspects of the problem, but differ in scope and methodology from our approach.

ASPECT : One of the most notable and advanced solver with excellent mesh adaptivity and numerics, and can simulate partial melting phenomenon. It follows the discussion in Schubert et al. (2001) and solves a Stokes system describing the movement of solid mantle material coupled with a advection-diffusion-reaction equation system transporting energy and chemical species. Phase change is included as a right hand side of the transport equation as a part of the reaction term. “However, this will not consider melt extraction or any relative movement between melt and solid and there might be problems where the transport of melt is of interest”.Bangerth et al. (2018), which, on the contrary, it is the main focus of our solver.

CitcomS : Zhong et al. (2000), Tan et al. (2006) A classical code used in simulation of mantle convection. It solves an incompressible Stokes problem coupled with energy transport, in consequence, melt extraction and chemical species transport are not well represented.

TerraFERMA : Wilson et al. (2017) A high level wrapper for FEniCS, PETSc and a science neural options handling system SPuD. The main focus of the code is to provide a rapid and reproducible construction of multi-physics model arising in geophysics scenarios.

pMELTS/ MELTS: The MELTS software is designed to investigate phase relations in magmatic systems during melting and crystallization. However, mechanical behaviors and transport not included.

1.3 Outline

The rest of the dissertation is organized as follows.

- In Chapter ??, we present a detailed description of the mathematical model we used to describe the partial melting process: mechanical flow, transport phenomenon and phase transition. In Section ??, We begin by deriving a Darcy-Stokes coupled system incorporating compaction equation from basic conservation laws. In Section ??, we describe transport of chemical species and energy within partial molten mantle. In Section ??, we present a simple linear parametrized two-component eutectic phase diagram. In Section ??, we nondimensionalize the physical model with carefully selected characteristic physical scales and formally present the nondimensionalized equations.
- In Chapter ??, we describe one new inf-sup stable MFEM solver equipped with serendipity function space and one new ML-WENO solver used to address the physical model defined previously. In Section ??, we describe

Chapter 2: Coupled Simulation

Now we couple locally mass conserved MFEM and FVM method together in this chapter and perform coupled numerical simulation for partially molten mantle. We discuss implementation details and coupling strategy combining MFEM, FVM and eutectic phase package. The remaining uncertainty of porosity ϕ will be fulfilled by phase package evaluation. Before we throw ourselves into the fully coupled problem. We test another porosity evolution scenario with a more straightforward prescribed melting. Then we will show an example combining MFEM, FVM and eutectic phase package altogether and perform coupled computation. We start with pseudo 1D scenarios and then proceed to perform full 2D simulation.

2.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ be the cell-averaged solution to the transport system (??) at time t^n . Let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ denote the solution to the Darcy-Stokes system at the same time step t^n . Moreover, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n given the transport result U^n .

For each time step t^n , we reconstruct dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ with ML-WENO reconstruction at each data points. The eutectic phase package gives corresponding phase parameters $\mathcal{W}^n$ according to these point wise values. Once we know $\mathcal{W}^n$, we solve a fixed in time Darcy-Stokes system at this time step t^n for $\mathcal{V}^n$. The cell-averaged solution $\mathcal{U}^{n+1}$ of the next time step t^{n+1} is then obtained by forward marching the system explicitly. We formally formulate an overall explicit coupled solver as shown in algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let U^0 be given as initial data.

- 1: Initialize $\mathcal{U}^0$, compute $\mathcal{W}^0$ with eutectic phase package.
- 2: Compute ϕ^0 in $\mathcal{W}^0$, solve the Darcy-Stokes system (??)–(??) for $\mathcal{V}^0$.
- 3: **for** Time level $n = 1, 2, \dots, n_{\max} - 1$ **do**
- 4: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.
- 5: Compute ϕ^n in $\mathcal{W}^n$, solve the Darcy-Stokes system (??)–(??) for $\mathcal{V}^{n+1}$.
- 6: Given $\mathcal{W}^n$ and $\mathcal{V}^{n+1}$, compute dimensionless effective velocity (??) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1 - \phi) \check{\mathbf{v}}_s}{\phi + (1 - \phi)D_i}, \quad \check{\mathbf{v}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

- 7: Explicitly marching forward the transport system (??) to next time $n + 1$ for $\mathcal{U}^{n+1}$.

8: **end for**

2.1.1 Discontinuous Data

Due to the nature of the finite volume method, we may obtain different reconstructed dimensionless enthalpy and mass concentration values from two sides of an element edge and thus two different phase calculation results. It is crucial to have one compatible and appropriate porosity value used on the edge for the continuous compaction equation in Darcy equation with divergence theorem involved in equation (??). We employ geometric average for porosity ϕ_e on an element edge e as

$$\phi_e = \frac{2\phi_e^+ \phi_e^-}{\phi_e^+ + \phi_e^-}, \quad (2.1)$$

where ϕ_e^+ and ϕ_e^- are porosity evaluated from two sides of the element edge e . When the porosity on one side equal zero, the geometric average of the porosity equals zero as well, which highlights the fact that there is no liquid flow travelling through this edge.

2.2 Pseudo One-Dimensional Column Tests

In this section, we first test our coupled algorithm on a 2D $M \times N$ grid with a pseudo-1D column problem. We test two different scenarios, one couples MFEM and FVM solver without phase computation but a simple prescribed melting function instead. The other one exhibits a fully coupled scenario where phase transition and porosity evolution present.

2.2.1 Prescribed Melting Case

Before fully coupled computation, we illustrate a simpler model coupling case coupling MFEM and FVM solver. Instead of deciding porosity with phase package computation, we prescribe the melting rate or the change of porosity as a fixed function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \quad (2.2)$$

We write conservation of liquid mass with the prescribed melting term Γ (2.2) as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \quad (2.3)$$

where diffusion effect is being dropped for simplicity.

We perform a quasi-1D column test on a 2×40 rectangular physical domain $\Omega \in [-0.05, 0.05] \times [-0.4, 0]$. We prescribe the following boundary conditions for MFEM and FVM solver.

- Bottom side;
 - MFEM. We prescribe solid velocity at the bottom of the physical domain, indicating supplying of mantle.

$$\tilde{\mathbf{v}}_s = \mathbf{v}_{\text{upwelling}}.$$

We prescribe zero liquid velocity indicating no melting penetrating bottom of the computational domain;

- FVM. We prescribe a fixed inflow boundary condition, supplying composition and enthalpy into system.
- Left and right side;
 - MFEM. We prescribe natural boundary condition giving zero traction to the Stokes equation in tangential direction. We prescribe zero flux for dofs in normal direction for the Stokes equation and Darcy equation.
 - FVM. We prescribe noflow (zero flux) condition.
- Top side;
 - MFEM. We prescribe natural boundary condition given zero traction and zero pressure to Stokes and Darcy part respectively.
 - FVM. We set free outflow boundary if it is recognized as outflow. We set zero if the velocity reversed to inflow boundary condition.

For simplicity and efficiency in computation, we use 1×3 and 1×2 for our ML-WENO(3,2) stencils, since we don't need resolution and high order accuracy in x direction. We show two different melting source scenarios. One of which is a piece wise constant stair function:

$$\Gamma_1(z) = \begin{cases} 1e-3, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \quad (2.4)$$

and one with melting source trapped in the middle of the melting column as

$$\Gamma_2(z) = \begin{cases} 0, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \quad (2.5)$$

We assign different initial conditions to these two scenarios as

$$\phi_1(z, 0) = 0.0, \quad z \in [-0.4, 0.0], \quad (2.6)$$

and

$$\phi_2(z, 0) = \begin{cases} 1e - 2, & -0.2 < z \leq 0, \\ 0, & -0.4 \leq z \leq -0.2. \end{cases} \quad (2.7)$$

2.2.2 Phase Transition Case

2.3 Two-Dimensional Test

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